

Introducing DRT and Person-Based Accessibility in MATSim

Jakob Rehmann^a, Kai Nagel^a

^a*Technische Universität Berlin, Chair of Transport Systems Planning and Transport Telematics, Germany*

Abstract

This work introduces two methodological developments in the MATSim accessibility contrib: Demand-Responsive Transport (DRT) accessibility and person-based accessibility. The DRT extension queries a running MATSim simulation for origin–destination- and time-specific level-of-service indicators, including waiting times and detour factors, and integrates these indicators into an econometric accessibility measure for pooled DRT. The person-based extension calculates accessibility from individuals’ home locations and allows sociodemographic characteristics to influence mode availability and travel utility. Together, the developments extend MATSim’s ability to analyse spatial and socioeconomic disparities in accessibility and the equity impacts of transport policies. Applications include the equity impacts of DRT in rural Germany and Japan, with a particular focus on older adults in the latter case, and the distributional impacts of the Berlin “Autofrei” proposal across income groups.

Keywords: accessibility, Demand-Responsive Transport, transport equity, person-based accessibility, rural mobility

1. Introduction

Accessibility measures are increasingly used to evaluate transport policies because they capture the potential of people to reach opportunities such as employment, education, shops, and health services. In conventional location-based approaches, accessibility is calculated from a set of geographic measuring points. This representation is useful for mapping spatial disparities, but it does not directly represent how accessibility differs between people living in the same area or how individual characteristics affect the transport options available to them. Demand-Responsive Transport (DRT) presents a related methodological challenge. Its level of service depends on the interaction between vehicle supply and customer demand, and therefore cannot be represented adequately by a single fixed travel time or a spatially homogeneous service parameter.

The work introduces two related extensions to the MATSim accessibility contrib. The first calculates accessibility for pooled DRT while preserving the service’s sensitivity to simulated supply and demand. The second extends accessibility calculations from geographic locations to individual persons. Both developments are implemented within the MATSim ecosystem, whose accessibility contribution supports utility-based calculations from transport networks, facilities, and time-dependent travel conditions. The methodological focus is on how these extensions work and how they can support equity-oriented policy analysis.

2. DRT accessibility

2.1. Methodological challenge

Pooled DRT combines the flexibility of individual transport with the shared-ride characteristics of public transport. Vehicles are not bound to fixed routes or timetables, but can serve several passengers during the same trip. Consequently, the travel time and waiting time experienced by a customer depend on the current fleet, vehicle locations, and booking requests from other passengers. These interactions make the service spatially and temporally heterogeneous.

An accessibility calculation requires travel characteristics between many origins and opportunities, often for several times of day. If all accessibility queries are submitted as normal DRT requests, the artificial demand generated by the calculation changes the simulated service and produces unrealistically long waits. If every query is simulated in isolation, the calculation removes the interaction between customers and eliminates pooling. The DRT extension addresses this problem by separating the extraction of service characteristics from the accessibility calculation.

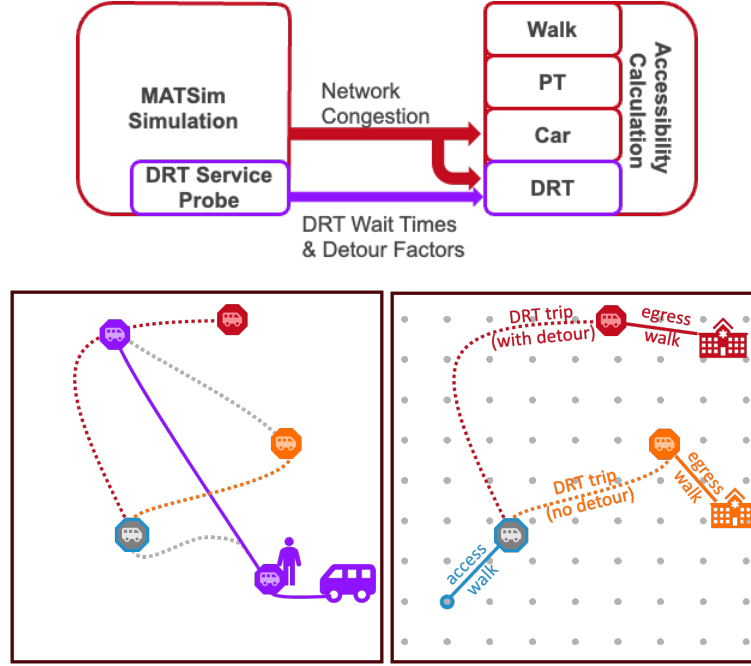


Figure 1: DRT accessibility methodology. Left: shows DRT service level probe. Purple trajectory displays a simulated DRT request, while orange and red trajectories show service probe requests (which will not be served). Right: shows econometric accessibility calculation. Gray dots denote measuring points and hospital symbols denote opportunities; accessibility combines access walking, the DRT leg, and egress walking.

2.2. Two-part DRT methodology

First, a running MATSim simulation is queried for an arbitrary set of origin–destination pairs and time slices. The DRT scheduling engine calculates a route and provides the relevant service characteristics, including waiting time and the detour factor associated with serving other passengers. The query is then discarded rather than fulfilled as a customer request. The probe therefore observes the service resulting from realistic supply–demand conditions without adding artificial demand to the simulation (see Fig. 1, left).

Second, the extracted level-of-service indicators are used in the accessibility calculation. For a measuring point ℓ and an opportunity k , a DRT trip can be represented as the combination of an access walk, a DRT leg between pickup and drop-off stops, and an egress walk:

$$V_{\ell k, \text{DRT}} = V_{\text{access}} + V_{\text{DRT}} + V_{\text{egress}} + \text{ASC}_{\text{DRT}}. \quad (1)$$

The DRT component includes the waiting time obtained from the simulation and a road-network travel time adjusted by the simulated detour factor. Utilities for alternative opportunities are aggregated using the existing logsum econometric accessibility formulation in MATSim. This enables pooled DRT to be compared consistently with car, public transport, walking, and cycling, while retaining the spatial and temporal variation in DRT service quality (see Fig. 1, right).

3. Person-based accessibility

The second development (see Fig. 2) changes the unit of analysis from a geographic measuring point to an individual person. In the location-based formulation, a point on a spatial grid is associated with a set of transport modes and a common accessibility value. In the person-based formulation, each agent’s home location becomes the measuring point, and the accessibility calculation is evaluated using that agent’s characteristics.

The framework uses MATSim’s modular routing and scoring components. An individual’s characteristics—e.g., age, car availability, and driving-licence status—can determine whether particular modes are available, the time and

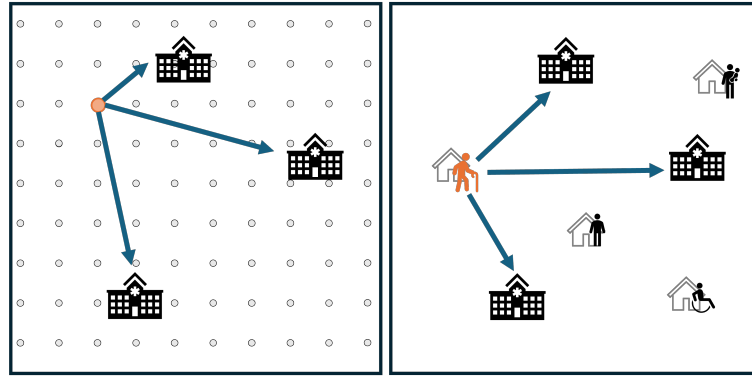


Figure 2: Location-based (left) vs. person-based accessibility (right)

distance needed to complete the route (e.g. elderly people might walk slower), and how much (dis)utility the agent receives through the travel.

Person-based accessibility is especially useful for equity analysis because it connects an accessibility outcome to the person who experiences it. It can therefore distinguish, for example, between older and younger residents, people with and without a driving licence, or households with different levels of car availability, even when they live in similar geographic locations.

4. Applications and relevance for transport equity

The methodological developments have been applied in three policy contexts. In the rural district of Kelheim, Germany, the DRT methodology is used to assess how a pooled service changes accessibility relative to public transport and how improvements are distributed between population groups and settlement types. The analysis demonstrates the relevance of combining spatially and temporally heterogeneous DRT service quality with an equity-oriented accessibility measure.

In Gunma Prefecture, Japan, the person-based framework is used to examine supermarket accessibility for residents with different ages, genders, home locations, and driving-licence statuses. The analysis focuses particularly on older adults, who are vulnerable to transport exclusion when they stop driving. The DRT scenarios illustrate how a flexible service can affect disparities between sociodemographic groups.

Finally, the person-based accessibility framework is used to examine the distributional impacts of the “Berlin Aut-ofrei” proposal across income groups, finding it particularly impacts wealthier residents. Across these applications, the methodological contribution is to provide a common simulation-based framework for comparing transport policies in terms of both accessibility and distribution. The framework can represent changes in transport supply, service quality, mode availability, and individual utility within the same MATSim environment. This makes it possible to move beyond average accessibility changes and examine who benefits, who loses, and how the results depend on the assumptions made about transport services and individual behaviour.